MILITARY SPECIFICATION

ALTIMETER, STANDBY, TYPES I AND II

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the performance, design, and test requirements for Standby Altimeters Types I and II. These altimeters are counter pointer integrally lighted with 2-inch dial for display of pressure-altitude for the range of -1,000 to 50,000 feet or -1,000 to 70,000 feet as applicable to type. The altimeter utilizes an electrical output which varies as a function of the barometric pressure setting.

1.2 Classification. Two types of altimeters are covered by this specification:

	Type	Range	Lighting	Potentiometer	Baro. output function
a.	I	-1,000 to 50,000 ft.	White (IPL)	10,000 ohms	in Hg.
b.	II	-1,000 to 70,000 ft.	White	5,000 ohms	Feet Altitude

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Boxes, Fiberboard

Military

MIL-P-116	Preservation-Packaging, Methods of
DOD-D-1000/1	Drawings, Engineering and Associated Data
MIL-B-5087	Bonding, Electrical, and Lighting Protection for Aerospace Systems
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum Alloys and Aluminum
MIL-C-14806	Coating, Reflection Reducing for Instrument Cover Glasses and Lighting Wedges
MIL-T-18303	Test Procedures, Production, Acceptance and Life for Aircraft Electronic Equipments, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-P-23408	Plating, Tin-cadmium (Electrodeposited)
MIL-C-25050	Colors, Aeronautical Lights and Lighting Equipment, General Requirements for
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for

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MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-B-27497	Bearing, Jewel, Sapphire or Ruby, Synthetic
MIL-C-38999	Connectors, Electric, Circular, Miniature, High Density, Quick Disconnect, Environment Resistant, Removeable Crimp Contacts
MIL-R-39023	Resistors, Variable, Nonwire-Wound, Precision, General Specifications for
MIL-G-81704	Glass, Aircraft Instruments, Lighting Wedges and Cover

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-147	Palletized Unit Loads (40" x 48" 4 Way Partial and 4 Way Pallets)
MIL-STD-454	Standard, General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-749	Preparation and Submission of Data for Approval of Non-Standard Electronic Parts
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipments Development and Production

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MIL-STD-794	Parts and Equipment, Procedure for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-859	Standard Calibration Table for Aeronautical Pressure Measuring Equipment

DRAWINGS

MS27467	Connector, Plug, Electric, Straight, Crimp Type, Bayonet Coupling
MS27560	2 3/8 Square Instrument, Standard Dimension for
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33649	Bosses, Fluid Connection Internal Straight Thread

2.1.2 Other Government publications. The following other Government publication forms a part of this specification to the extend specified herein.

Department of Defense Standardization

SD-6	Provisions Governing Qualification

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

3.1 Qualification. The standby altimeters furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.4 and 6.4).

3.2 Parts and materials.

3.2.1 General. Parts and materials shall meet the requirements of MIL-E-5400 for Class 2 equipment.

3.2.1.1 Nonstandard parts. Standard MS and AN parts shall be used where they suit the purpose. When no standard part is available, a nonstandard part may be used with prior approval of the procuring activity (see 6.2.1). Selection and approval of the use of nonstandard parts and materials shall be obtained by the contractor as outlined in MIL-STD-749.

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3.2.1.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2.2 Materials.

3.2.2.1 Metals. Metals shall be of the corrosion-resistant type unless suitably protected to resist corrosion during the life of the instrument.

3.2.2.2 Nonferrous materials. Nonferrous materials shall be used for all parts of the altimeter except where ferrous materials are essential.

3.2.2.3 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts except where magnetic materials are essential. The extent to which magnetic materials are required shall be reported to the procuring activity (see 6.2.2).

3.2.2.4 Dissimilar metals. Unless protected against electrolytic corrosion by means of a protective coating, intimate contact of dissimilar metals shall be avoided. Dissimilar metals are defined in MIL-STD-454, requirement 16.

3.2.2.5 Protective treatment. When materials used in the construction of the altimeter are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service life, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.2.2.6 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with fungicidal agent acceptable to the procuring activity (see 6.2.2).

3.2.2.7 Toxic and corrosive fumes. The materials, as installed in the altimeter and under the conditions specified herein, shall not liberate deleterious or corrosive fumes. This shall include any fungicidal agents that are used.

3.2.2.8 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except that the dial, small holes, case inserts, and fine pitch gear teeth need not be anodized. Parts which do not anodize satisfactorily may be coated in accordance with MIL-C-5541.

3.2.2.9 Components. The electronic and mechanical components shall be in accordance with MIL-E-5400.

3.2.2.10 Jewel bearings. All sapphire and ruby jewel bearings used in the altimeter shall be in accordance with MIL-B-27497.

3.2.2.11 Threads. The threads of all machine screws shall be in accordance with MIL-S-7742.

3.3 Design and construction. The altimeter shall conform with all applicable requirements of MIL-E-5400 concerning design, construction, and workmanship. The design shall include minimum size and weight consistent with maximum reliability

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and operability. In any case, it is a condition of final acceptance that the equipment manufactured shall meet all the requirements of this specification, and any applicable acceptable test documents.

3.3.1 Physical characteristics.

3.3.1.1 Description. The altimeter shall consist of a precision all-mechanical altimeter mechanism, a counter pointer type altitude display, integral red or white lighting, a barometric setting displayed by means of a four digit counter, and a potentiometer whose output varies as a function of the barometric setting. All principle dimensions and locations of particular features shall be as detailed in table I and figure 1.

TABLE I. Weight and dimensions for the altimeter.

Parameters	Value (Maximum)	
1/ Length	7.124 inches	18.09 centimeters
Width	2.385 inches	6.058 centimeters
Height	2.385 inches	6.058 centimeters
2/ Weight	1.5 pounds	0.680 kilograms

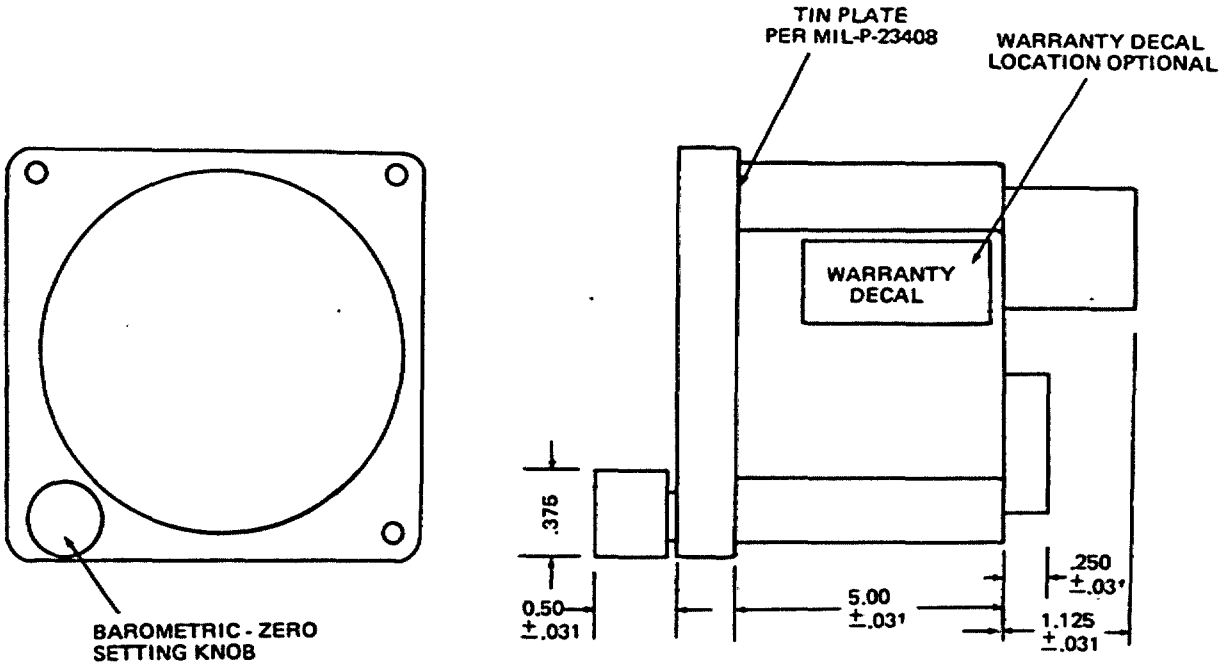
1/ Measurements include control knob and connector.
2/ Without fittings and mounting screws.

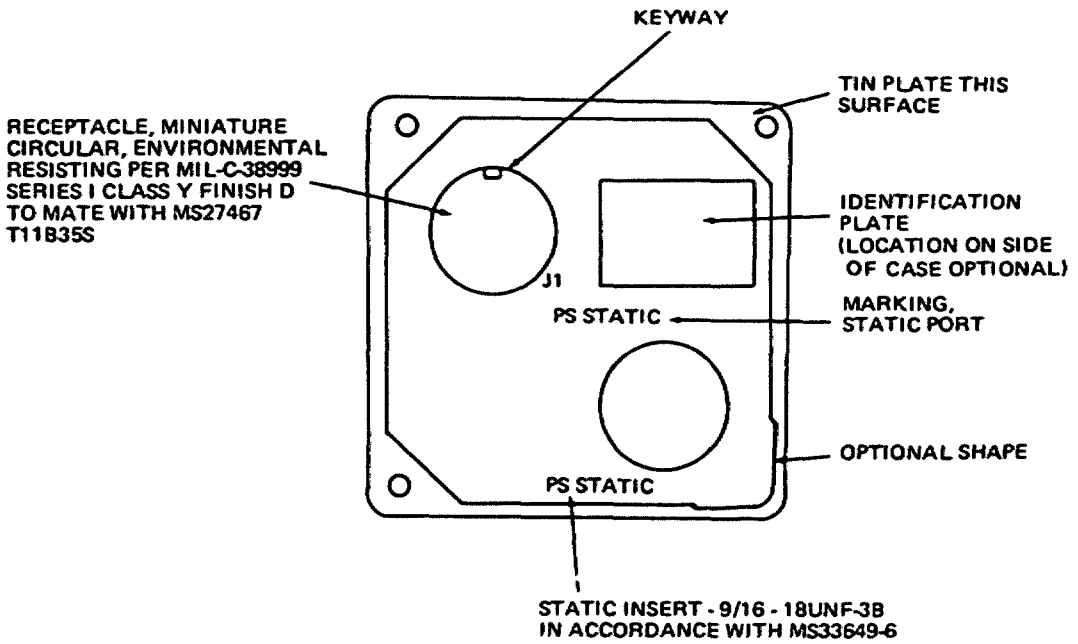
3.3.1.2 Case. The outline dimensions and the external connections of the altimeter case shall conform to figure 1. The body and mounting flange of the case shall be made of lightweight non-magnetic metal, uniform in texture. The body shall be finished in a durable black finish, and the mounting flange shall be finished with a durable lusterless black finish, Color No. 37038 of FED-STD-595. The back of the mounting flange shall be tin plated in accordance with MIL-P-23408 to allow electrical grounding to the aircraft instrument panel and to comply with MIL-B-5087 requirements.

3.3.1.3 Static pressure connection. The static connection shall be in accordance with figure 1 and shall be located as shown thereon. A 150-200 wire mesh filter screen (either stainless steel or monel) shall be firmly installed in the static connection to prevent dirt particles from entering the case. For storage, shipping, and handling purposes, a vented plug of acceptable material and design shall be provided. The plug shall incorporate a vent consisting of a hole at least 1/16 inch in diameter. The plug shall also incorporate suitable filtering material to prevent the entry of particles which the manufacturer considers would be detrimental to the altimeter's performance.

3.3.2 Reliability. The contractor shall establish a reliability assurance test program in accordance with MIL-STD-785.

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NOTES:

1. THE ELECTRICAL CONNECTOR SHALL BE LOCATED IN THE UPPER LEFT QUADRANT OF THE UNIT AND THE STATIC CONNECTION SHALL BE LOCATED IN THE LOWER RIGHT QUADRANT OF UNIT.
2. THE CASE SHALL BE MANUFACTURED IN ACCORDANCE WITH MS27560 EXCEPT AS INDICATED.
3. TOLERANCES SHALL BE 0.010 INCHES UNLESS OTHERWISE NOTED.

FIGURE 1. Altimeter outline

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3.3.2.1 Operational stability. The altimeter shall meet specified performance requirements when operated continuously or intermittently, for a period of at least 1000 hours. During this time, it shall not be necessary to readjust any controls that are inaccessible to the operator during normal use.

3.3.2.2 Operating life. The altimeter shall have a minimum operating life of 10,000 hours with reasonable servicing and replacement of parts.

3.3.2.3 Reliability in mean-time-between-failures (MTBF). The altimeter shall have a specified mean (operating) time between failures of 1000 hours when tested as specified in 4.6.31.2.

3.3.3 Cabling and connections.

3.3.3.1 Electrical receptacle. The electrical receptacle, used to interface with aircraft wiring, shall conform to specification MIL-C-38999 and shall be as specified in figure 1. Figure 2 shows receptacle wiring for aircraft interface. The electrical receptacle shall be protected with a suitable cap for shipping, storage, and handling.

3.3.3.2 Case grounding. The case shall be internally grounded as shown in figure 2. The electrical system in the altimeter shall not be grounded.

3.3.3.3 Interconnect cabling. The altimeter shall be capable of satisfactory performance using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall not be shielded. However, the altimeter shall incorporate electrical receptacle pins grounded inside the case adjacent to the receptacle in the event that shielding is necessary for specific aircraft installations. The pins as designated in figure 2 shall be provided for this purpose.

3.3.4 Interchangeability. Interchangeability shall exist between all units and replaceable assemblies, subassemblies, and parts for all equipment delivered on the contract in accordance with MIL-STD-454, requirement 7. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of DOD-D-1000/1.

3.3.5 Interference control. The generation of electromagnetic interference by the altimeter and the vulnerability of the altimeter to electromagnetic interference shall be controlled within the limits of MIL-STD-461 for Class A1b equipment. EMI testing shall be accomplished under standard conditions specified in 3.3.10 and in accordance with the procedures of MIL-STD-462 to demonstrate compliance with the requirements of MIL-STD-461.

3.3.6 Maintainability. The altimeter shall be repaired at depot level maintenance shops only. Maintainability shall be demonstrated in accordance with MIL-STD-471. An equivalent plan to demonstrate maintainability prior to the performance of qualification tests may be used upon approval of the qualifying activity (see 6.2.1).

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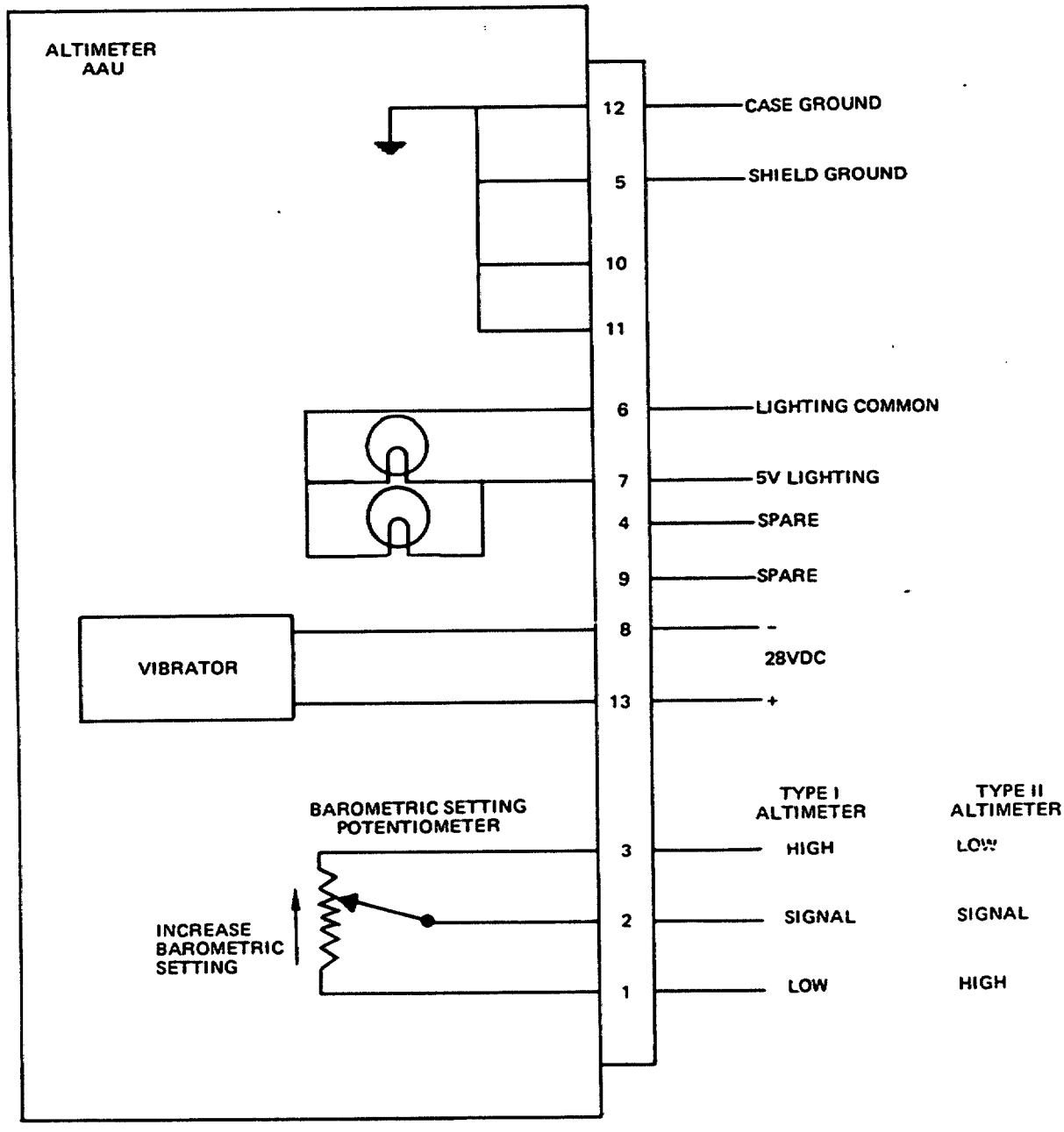


FIGURE 2. Altimeter electrical wiring diagram

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3.3.6.1 Repair time. The altimeter shall be completely repairable at depot level. Repair time for the altimeter shall be not greater than 8 hours using one depot technician and authorized depot AGE and technical information. Spare parts and facilities are assumed to be available. Repair time shall include all repair and test time excluding government or manufacturer acceptance testing.

3.3.7 Transportability. The altimeter shall be transportable by air, rail, truck, or ship when packaged as specified in section 5.

3.3.8 Shelf life. The altimeter shall meet all performance requirements after storage or inactivity for a period of thirty days. The normal storage environmental extremes shall be any dew point between -62°C and +95°C.

3.3.9 Nameplate and identification marking. Serial number assignment, nameplate approval, and identification marking shall be in accordance with MIL-N-18307.

a. Unit name (Altimeter, Standby, Type I, Type II)

b. Unit nomenclature

c. Contract number

d. Manufacturer's code number

e. Equipment drawing number

f. Procuring activity (Department of Navy-Naval Air Systems Command)

g. Serial number

3.3.9.1 Decals, nameplates, and marking. All decals, nameplates, and marking shall be affixed to the altimeter in accordance with figure 1 and all referenced marking specifications.

3.3.9.2 Serial numbers. Serial numbers will be assigned by the procuring activity for equipment under contract or order. The contractor shall serialize sequentially all equipments which are delivered under the contract.

3.3.10 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench test.

a.	Temperature	Room ambient (25 ±5°C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient 40 to 90 percent relative humidity

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e.	Power	5.0 ±1.0 volt AC or DC, 4.0 watt maximum 28 ±1.0 volt DC, 90 ma 30 ±1.0 volt DC, 6.9 ma (Type II altimeter) 10 ±1.0 volt DC, 1.2 ma (Type I altimeter)

3.3.11 Service conditions. The altimeter shall meet the specified performance requirements when operated under the following environmental service conditions.

3.3.11.1 Temperature.

3.3.11.1.1 Low temperature. The altimeter shall meet the operational requirements of this specification while being subjected to low temperature tests in accordance with 4.6.27.1.1.

3.3.11.1.2 High temperature. The altimeter shall meet the operational requirements of this specification while being subjected to high temperature tests in accordance with 4.6.27.1.2.

3.3.11.2 Vibration. The altimeter shall meet the operational requirements of this specification while being subjected to nongunfire and gunfire vibration exposure in accordance with MIL-STD-810, method 514.2 and method 519.2, respectively, except as modified herein.

3.3.11.3 Acceleration. The altimeter shall meet the operational requirements of this specification while being subjected to the acceleration tests for carrier based aircraft in accordance with MIL-STD-810, procedure II. The altimeter shall not be physically damaged when subjected to acceleration tests for carrier based aircraft in accordance with MIL-STD-810, procedure I.

3.3.11.4 Humidity. The altimeter shall operate satisfactorily after being subjected to a humidity test in accordance with MIL-STD-810, method 507.1, procedure I. The altimeter shall not be damaged as a result of the humidity test.

3.3.11.5 Fungus. The altimeter shall not be adversely affected by fungi when subjected to fungus tests in accordance with MIL-STD-810, method 508.2.

3.3.11.6 Salt spray. The altimeter shall meet the operational requirements of this specification immediately after, and again 48 hours after, salt spray exposure in accordance with MIL-STD-810, method 509.1, procedure I.

3.3.11.7 Rain. The altimeter shall meet the operational requirements of this specification after exposure to rain tests in accordance with MIL-STD-810, method 506.1, procedure I and 4.6.27.7.

3.3.11.8 Sand and dust. The altimeter shall meet the requirements of this specification when subjected to sand and dust tests in accordance with MIL-STD-810, method 510.1, procedure I.

3.3.11.9 Shock.

3.3.11.9.1 Basic design shock test. The altimeter shall meet the operational requirements of this specification during and following basic design shock tests in accordance with MIL-STD-810, method 516.2, procedure I, and 4.6.27.9.1.

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3.3.11.9.2 Crash safety shock test. The altimeter mounting attachment shall be permitted to bend and distort when subjected to crash safety shock test in accordance with MIL-STD-810, method 516.2, procedure III and 4.6.27.9.2. However, the mounting attachment shall not break as a result of this test.

3.3.11.10 Rates of climb and descent. The altimeter shall provide accurate altitude indications at change of altitude of up to 30,000 feet per minute (fpm). The altimeter shall sustain no damage at change of altitude of 50,000 fpm for type I and 80,000 fpm for type II.

3.3.11.11 Dielectric strength. The altimeter shall withstand the application of 500 volts AC at a frequency of 60 Hz for 30 seconds.

3.3.11.12 Magnetic property. The altimeter shall not cause a free compass to deflect more than 3 degrees when located 5 1/2 inches from the center of the compass.

3.3.11.13 Case pressurization. The altimeter shall not be damaged in any way and shall perform satisfactorily after exposure to an external pressure of 60 in. Hg. absolute for 4 hours with the static port connected to ambient pressure.

3.3.11.14 Case leakage. The case leakage shall be not greater than the equivalent of an altitude change of 100 feet per minute when subjected to an altitude exposure of 40,000 feet.

3.3.11.15 Overpressure. The altimeter shall not shift zero or calibration when subjected to a pressure of 35 in. Hg. (absolute).

3.3.11.16 Underpressure. The type I altimeter shall not shift zero or calibration when subjected to altitude exposure of 60,000 feet. The type II altimeter shall not shift zero or calibration when subjected to altitude exposure of 100,000 feet.

3.3.11.17 Scale error. The altimeter shall indicate altitude within the accuracy requirements of table II.

3.3.11.18 Friction error. The friction error tolerance shall be not greater than the values specified in of tables III and IV.

3.3.11.19 Position error. Position error shall be not greater than 20 feet when altimeter is subjected to the test in accordance with 4.6.8.

3.3.11.20 Hysteresis. Hysteresis shall be not greater than 40 feet at sea level after the altimeter has been exposed to a pressure altitude of 50,000 feet.

3.3.12 Service conditions (nonoperating). Service conditions for shelf life and transportation shall meet the shelf life and transit requirements of 3.3.7 and 3.3.8.

3.3.12.1 Warm-up time. The time required for the equipment to warm-up prior to operations shall be not greater than 2 minutes either under standard conditions or at extreme service conditions.

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TABLE II. Scale errors.

Equivalent Pressure In. Hg. at 0°C	Standard Altitude Feet	Scale Error Tolerances (Feet)		
		-35°C	Room Temp. and +45°C	-54°C and +71°C
31.018	-1000	±60	±40	
1/ 29.921	0	60	40	±80
29.385	500		40	
28.856	1000	60	40	80
1/ 27.821	2000	80	50	100
26.817	3000	95	55	120
1/ 24.896	5000	110	65	140
22.224	8000		85	
1/ 20.577	10000	150	95	190
19.029	12000		105	
16.886	15000	190	120	240
14.942	18000		135	
1/ 13.750	20000	270	155	320
11.103	25000	300	180	360
8.885	30000	330	205	400
1/ 7.041	35000	360	230	450
1/ 5.538	40000	390	255	500
4.355	45000	430	300	580
3.425	2/ 50000	600	450	800
2.693	2/ 55000	900	600	1200
1/ 2.118	2/ 60000	1200	800	1600
1.310	2/ 70000	1800	1200	2400

NOTES:

1/ Apply for environmental and reliability tests as specified herein.
2/ Apply to type II altimeters only.

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TABLE III. Stop and jump friction error.

Column "A" Test Point Altitude (Feet)	Column "B" Rate of Ascent or Descent (Feet/Minute)	Column "C" Tolerance Vibrator Operative (Feet)	Column "D" Tolerance Vibrator Inoperative	
			3/ Initial (Feet)	4/ Final (Feet)
1/ 0	500	25	-	-
1,000	500	25	150	200
3,000	500	25	150	225
1/ 5,000	500	25	150	250
1/ 10,000	500	25	400	400
20,000	3,000	25	400	575
1/ 30,000	3,000	25	400	800
35,000	3,000	50	600	
1/ 40,000	3,000	50	600	1200
2/ 50,000	3,000	50	750	
1/2/ 60,000	3,000	50	1000	
2/ 70,000	3,000	75	1500	

NOTES: 1/ Apply for environmental and reliability tests as specified herein.
2/ Apply to type II altimeters only.
3/ "Initial" tolerances shown apply to all friction tests.
4/ Upon completion of reliability tests and upon completion of vibration endurance tests, "Final" tolerances are applicable.

TABLE IV. Static friction error-vibration inoperative.

Test Point Altitude (Feet)	Tolerance	
	2/ Initial (Feet)	3/ Final (Feet)
1/ 500	100	125
1,500	100	125
2,500	100	125
3,500	100	125
1/ 7,500	100	175
12,500	100	175
17,500	110	200
22,500	120	250
1/ 32,500	180	300
35,500	300	
1/ 42,500	300	400

1/ Apply for environmental and reliability tests.
2/ "Initial" tolerances shown apply to all friction tests.
3/ Upon completion of reliability and vibration endurance test, "final" tolerances are applicable.

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3.3.13 Input electrical power.

3.3.13.1 Operating power. The altimeter shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall be not greater than the specified amounts.

a. Lighting power: 5.0 ±0.1 volt AC or DC, 4.0 watt maximum.

b. Integral vibrator power: 28 ±1.0 volt DC, 90 ma.

c. Potentiometer (Type I Altimeter): 30 ±1 volt DC, 1.2 ma.

d. Potentiometer (Type II Altimeter): 30 ±1 volt DC, 6.9 ma.

3.3.13.2 Overload protection. All parts and circuits of the altimeter which are likely to carry an overload due to any failures or poor adjustments shall be proportioned to withstand such overload without permanent damage. The use of fuses and other protective devices shall not be permitted.

3.3.13.3 Undervoltage protection. The altimeter shall not be damaged by voltage below the minimum specified herein and shall automatically resume normal operation when the voltage is within the specified limits.

3.4 Performance. The performance of the altimeter shall be as set forth below. Unless otherwise specified herein, these requirements shall apply under both standard and service conditions. When reduced performance requirements for service conditions are acceptable, such variations are specified herein.

3.4.1 Barometric setting electrical output, type I altimeter. The electrical output of the barometric scale setting shall be provided by means of a potentiometer. The total resistance of the potentiometer shall be 10K ±1K ohms. The potentiometer shall be connected to the electrical receptacle pins as shown in figure 2. When the wiper is loaded by 1 megohm, the measured resistance ratio shall increase linearly with barometric setting. The wiper shall cover 90 percent of the total resistance for pressure settings from 28.10 to 31.00 inches of mercury (in. Hg.). The design resistance ratios for various pressure settings and the applicable accuracy requirements are listed in table V. The potentiometer shall be electrically isolated from the altimeter and shall not exhibit potentiometer noise greater than specified by MIL-R-39023. The output resistance ratio shall be repeatable within ±0.00093 when a specific barometric pressure setting is made from a clockwise or counter-clockwise direction. Operation of the barometric setting mechanism below 28.10 in. Hg. or over 31.00 in. Hg. shall not result in an output resistance ratio below 0.035 or above 0.965, respectively.

3.4.2 Barometric setting electrical output, type II altimeter. A potentiometer output shall be provided such that an increase in the setting of the barometric pressure will result in a decrease in the potentiometer electrical output voltage. The output voltage change shall be linear with the equivalent altitude represented by the increase in setting. The total resistance of the potentiometer shall be 5000 ±500 ohms. When 30 volts dc is applied across the potentiometer, the output voltage as measured from the wiper to 30V DC (low) through a 1 megohm loading resistor

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TABLE V. Barometric setting scale.

Barometric Pressure Scale Setting (Inches Hg.)	Altitude Display Correct Difference (Feet)	Type I Altimeter		Type II Altimeter	
		Resistance Ratio 1/ 2/	Ratio Tolerance	Voltage (Volts) 4/ 5/	Tolerance (Volts)
28.10	-1727	0.05000	± .003	27.27	± 0.25
28.50	-1340	0.17414		23.40	± 0.25
28.86	-996	0.28586		19.96	± 0.25
3/ 29.00	-863	0.32931	.002	18.63	± 0.25
29.31	-570	0.42552		15.70	± 0.25
29.50	-392	0.48448	.0015	13.92	± 0.25
3/ 29.92	0	0.61483	.0015	10.00	± 0.25
30.03	+100	0.64897	.0015	9.00	± 0.25
30.50	+531	0.79483	.0015	4.69	± 0.25
30.73	+740	0.86621		2.60	± 0.25
3/ 30.90	+893	0.91897		1.07	± 0.25
31.00	+983	0.95000	.003	0.17	+ 0.25

1/ Independent Linearity: Over any 0.30 in. Hg. output function segment the maximum output deviation from the best straight line drawn through the output function data points within the selected segment shall not exceed 0.002 resistance ratio.

2/ Repeatability: ± 0.0093 resistance ratio (equivalent to 0.003 in. Hg.)

3/ Applies to environmental test

4/ Any settable pressure setting below 28.10 shall not cause the voltage from wiper to ground to exceed 29 volts.

5/ Repeatability: ± 0.20 volts (equivalent to ±20 ft. altitude)

shall decrease by 1 volt for each 100 feet of change in altitude. The voltage output shall be 10 volts at the standard sea level barometric pressure setting 29.92 in. Hg. The required voltages for various pressure settings are listed in table V. Any pressure setting below 28.10 in. Hg. shall not result in an output voltage greater than 29 volts. Any pressure setting above 31.00 in. Hg. shall not result in an output voltage lower than 0.1
7 volts. The potentiometer shall be connected to the altimeter as shown in figure 2.

3.4.3 Pointer and counter relationship. The pointer shall indicate readings in feet with reference to a single uniformly graduated scale, making one revolution for each 1,000 feet of altitude. The counter shall be viewed through an aperture in the dial and shall indicate in the thousands of feet throughout the range. During ascent when the pointer passes the 0 mark, the counter shall have just changed its indication by one digit in such a way that the possibility of misreading the indication is minimized. During descent, when the pointer passes the 0 mark, the counter shall start to change its indication by one digit in such a way that the possibility of misreading the indicator is minimized. The counter transition shall be accomplished within the short period when the pointer is moving between 9 and 0 on

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the dial. During the long period when the pointer is moving between 0 and 9, the 1,000 feet drum shall be locked in position and in alignment with 10,000 feet drum. Each digit on the 1,000 feet drum shall represent 1,000 feet intervals. The 10,000 feet drum shall be driven by the 1,000 feet drum so that the 10,000 feet drum will index one digit as the 1,000 feet drum moves through the short period between 9 and 0 in either direction. During the long period between 0 and 9 of the 1,000 feet drum, the 10,000 feet drum shall be locked in position. Each digit of the 10,000 feet drum shall represent 10,000 feet levels. In the space corresponding to zero, the 10,000 feet drum shall have striped lines as shown in figure 3. The 1,000 feet drum and the 10,000 feet drum are collectively called the counter. The counter weight of the pointer shall not in any position obscure the indication of the drum. The combined readings of the two counter drums shall indicate the pressure altitude in thousands of feet. The pointer shall act as a vernier of the thousands drum, as well as being an indication of trend information. The rotation of the altitude drums shall be such that the higher numbers appear from the top of the display with increasing altitude.

3.4.3.1 Pointer. The pointer shall be free to swing to either side of the zero position without engaging a mechanical stop. The pointer shall be firmly attached to the pointer shaft to prevent slippage under normal service use and still be removable for depot service maintenance. The pointer shall not distort and bind as a result of application of temperature differentials such as could result from sunlight on the dial or other service conditions.

3.4.4 Barometric setting system. A barometric setting system, controlled by a knob, shall be provided that will permit the altimeter display to be set to indicate zero altitude at any existing ground level pressure throughout the range of 28.10 to 31.00 (inclusive) inches of mercury in 0.01 inch increments. The gear ratio between the setting knob and the display pointer shall be such that one complete revolution of the setting knob shall result in a display pointer movement of approximately 200 to 300 feet. Clockwise rotation of the setting knob shall result in clockwise rotation of the pointer and the higher numbers shall appear from the top of the barometric setting counters. The shaft of the setting knob shall be capable of withstanding a load of 15 pounds applied perpendicular to the shaft at the extreme outward end of the knob without permanent deformation. A positive stop or stops shall be provided to limit the barometric setting mechanism movement such that the setting range of 28.0 to 31.1 inches of mercury is not greatly exceeded. The setting system shall not introduce errors in the altitude indication greater than specification tolerances when set at any point within the range established by the stops. The stops shall be capable of withstanding a torque of 6 pound inches applied to the barometric setting knob without damage to the stops or the mechanism. If the design is such that the stops or the mechanism may be damaged by the application of 6 pound inches torque to the knob, the use of a torque limiting mechanism incorporated in the knob or within the altimeter is acceptable if specification requirements are otherwise met. Torque limiting devices which may slip during normal adjustment or which may otherwise malfunction are not acceptable. The design of such a device is subject to the approval of the qualifying agency. The resistance (torque) when setting the barometric pressure shall be within the range of 2 to 7 ounce inches. The barometric counter unit displaying the setting shall be located as shown in figure 3. The barometric setting mechanism shall be capable of withstanding a push and a pull of 15 pounds while the barometric scale is rotated through the range of setting from the minimum stop to the maximum stop without disengagement or damage of

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the mechanism. The barometric setting system shall be designed to prevent the barometric setting counter display and the electrical output of the barometric pressure setting from shifting or changing as a result of exposure to vibration within the range described in this specification.

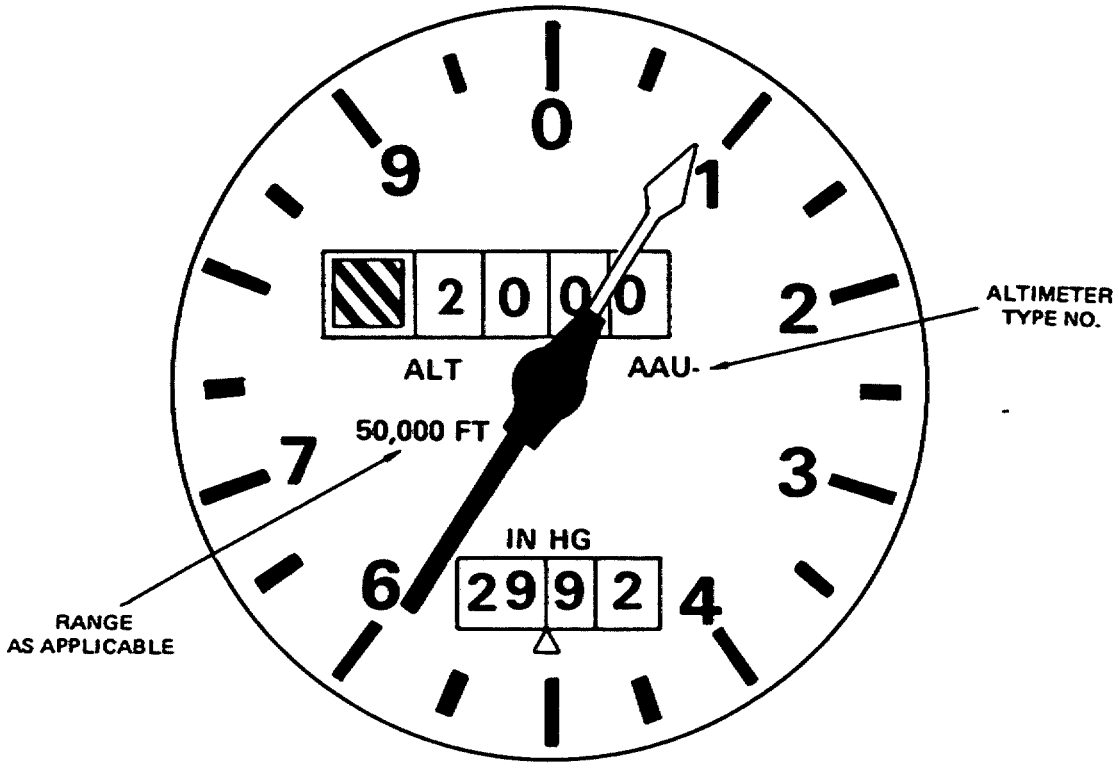
3.4.4.1 Barometric setting knob. The barometric pressure setting knob shall be attached in such a manner that it can be easily removed from its shaft and replaced with a knob of different length. An attachment means, equivalent to a flattened surface on the shaft, and a setscrew(s) shall be used to provide positive securing for turning.

3.4.4.2 Adjustment of barometric scale counter to pointer. The barometric setting system shall be designed to permit adjustment of the relative setting of the pointer and barometric scale counter. The barometric setting system shall be capable of correcting for zero shifts encountered in service use without requiring that the altimeter be returned to overhaul or to the manufacturer for the purpose. If the design is such that the relative setting of the barometric scale counters and the barometric potentiometer output is affected by this adjustment, a means must also be incorporated to adjust the output. The design of any potentiometer output adjustment shall provide for setting from the face of the altimeter when it is installed in a panel. The design shall be approved by the qualifying activity. The barometric scale counter to pointer setting mechanism shall be provided with a lock which shall hold the setting knob in its normal position and allow the aircraft pilot to make adjustments as specified by 3.4.4. The means for unlocking the barometric setting systems shall be accessible from the face of the altimeter when installed on the instrument panel. The unlocked barometric setting system shall allow a technician to make the adjustment of the relationship between the barometric scale counter and the pointer should it be required. The design shall provide for a seal(s) to prevent unauthorized adjustment(s). The locking mechanism shall be designed to prevent the loss of the locking screw, locking pin cover, or any other part of the locking mechanism while rezeroing the altimeter. The design of the locking mechanism shall be such that in the event the locking device is accidentally loosened, movement of the barometric setting knob to the barometric scale adjustment position shall not be possible without further action such as pulling up on the locking screw to disengage the lock. It is intended that accidental adjustment of the relative settings of the pointer barometric scale counter or potentiometer output by the aircraft pilot is precluded in the event the locking device becomes loose.

3.4.5 Integral lighting. The altimeter shall be provided with integral lighting in accordance with MIL-L-25467 and MIL-L-27160 as applicable to illuminate the display. The lighting circuit shall operate from 5 VAC or DC and shall be connected in accordance with figure 2. The integral lighting wiring shall not be grounded to the altimeter case. The bulbs shall be readily replaceable without soldering unless it can be demonstrated by documentation and test that the lighting system has a minimum MTBF of 10,000 hours. The lighting colors of the altimeter shall be specified in MIL-C-25050 and in 1.2 of this specification. The design shall be such that red lighting or white lighting, as applicable, can be incorporated by means of a lamp filter change or lamp assembly change only.

3.4.5.1 Coverglass and lighting wedge. The coverglass and/or lighting wedges shall conform to MIL-G-81704 Type I and shall not interfere with the readability of the altimeter. All reflecting glass surfaces shall be provided with a reflection-reducing coating that meets the requirements of MIL-C-14806.

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NOTE: DETAIL DIMENSIONS SHALL BE AS SPECIFIED IN 3.7.

△INDICATES LOCATION OF DECIMAL POINT.

FIGURE 3. Altimeter dial

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3.4.5.2 Distance from dial to coverglass. The distance from the dial to the innermost glass surface shall be not greater than 0.250 inch.

3.4.6 Internal vibrator. An internal vibrator shall be incorporated into the altimeter and shall effectively decrease the friction in the mechanism. The vibrator shall be self-starting upon being energized and shall start without being tapped or require other external vibration assistance, and shall provide adequate vibration to remove friction under all environmental conditions specified herein. The vibrator unit shall be capable of starting, running and producing vibration sufficient to meet the friction and reliability requirements specified herein. The design shall be such that extended use of the vibrator will not result in deposits of loose material nor produce deleterious fumes that may result in adverse effects on altimeter performance. The internal vibrator power shall be as specified in 3.3.13.1b. The vibrator circuit shall be capable of withstanding a reverse polarity of 28 volts dc without damage to the circuit.

3.5 Mechanical adjustments. The altimeter shall be provided with a means of adjusting or correcting the indications to conform to performance required herein, if adjustments should become necessary during the life of the altimeter. The calibration adjustment shall not affect the temperature performance of the altimeter. This means of adjustment shall be reasonably simple to manipulate and of such nature that adjustment can be accomplished by tools ordinarily used by an altimeter repairman. The adjustment shall be made in accordance with the manufacturer's instructions (Government approved) at the level of maintenance for which such an adjustment is authorized by the Government.

3.6 Dial. The dial shall be marked as shown in figure 3. Unless otherwise specified in the contract or purchase order, all visible portions of the dial shall be lusterless black Color No. 37038 of FED-STD-595 and all markings on the dial shall be lusterless white Color No. 37875 of FED-STD-595. The dimensions of the dial markings shall be as specified in 3.7. The dial shall be securely fastened to the mechanism frame by means of screws or mechanical locking and shall be of sufficient thickness that warping does not result when the altimeter is subjected to the environmental conditions specified herein. The dial shall be removable at overhaul without damage.

3.6.1 Visibility of dial. The pointer, numerals (at least 1/32 inch of the shortest graduations), and all other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 20 degrees with a perpendicular to the dial and whose small diameter is the aperture of the case.

3.6.2 Markings. All markings shall be durable so as to withstand the usage encountered in service. The style and proportions of numbers and letters placed on the dial shall conform to MS33558.

3.6.3 Numerals. Each number shall distinctly indicate the graduation to which it applies. If practicable, each number shall be so placed that the center mass of the number is on the radial line joining the appropriate graduation and the center of the dial.

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3.7 Finish and dimensions. The altimeter markings shall be finished in lusterless white, Color No. 37875 of FED-STD-595 or lusterless black, Color No. 37038 of FED-STD-595 as specified in table VI.

3.8 Workmanship. Workmanship for the altimeter shall conform with requirement 9 of MIL-STD-454.

TABLE VI. Marking dimensions.

Markings	Height or Length Inch ±0.015	Width of Line or Graduation Inch ±0.005	Color
Numbers 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, on counter and fixed zeros to the right of the counter	0.156 Min.	0.023	White
Numerals 0, 1, 2, 3, 4, 5, 6, 7, 9 on dial	0.187	0.023	White
100-foot graduation	0.125	0.020	White
50-foot graduation	0.072	0.010	White
Baro counter numerals	0.156	0.025	White
ALT lettering	0.156	0.025	White
IN HG lettering	0.046	0.008	White
100 FT lettering	0.046	0.008	White
1000 FT lettering	0.046	0.008	White
Shaded portion of pointer	-	-	White
Border on altitude counter window	-	0.035	White
50,000 FEET lettering	0.062	0.011	Black
70,000 FEET lettering	0.062	0.011	Black
Unshaded portion of pointer, dial, counter	-	-	Black
AAU-	0.062	0.011	Black

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The manufacturer shall ensure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions and precautions.

4.3.1 Conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the conditions specified in 3.3.10.

4.3.2 Test readings. Unless otherwise specified herein, vibrator and lighting power shall be applied to the altimeter during all tests and the barometric output potentiometer shall be electrically loaded to simulate operating conditions. The altimeter shall not require external vibration or tapping to meet the performance specified herein. Application of external vibration in the order of 0.2 g or tapping the altimeter shall not cause the altimeter to exceed the applicable tolerances.

4.3.3 Attitude. The altimeter shall be tested in its normal operating position.

4.3.4 Integral lighting. The altimeter lighting system shall be energized and shall operate satisfactorily during all tests when an operational requirement for the altimeter is specified. The lighting shall be tested and operate satisfactorily after completion of each non-operational test.

4.3.5 Rate of pressure change. Unless otherwise specified herein, the rate of change in pressure during all tests for both increasing and decreasing altitudes shall be accomplished at any rate up to 30,000 feet per minute. The rate shall be progressively reduced as the checkpoints are approached to avoid passing the checkpoint. The movement of the pointer during increasing and decreasing altitudes shall be smooth and free from irregular motion when the integral vibrator is operating.

4.3.6 Altimeter zero setting. Resetting of the altimeter shall not be accomplished at any time during the course of testing and shall not be required to meet the accuracy tolerances specified herein except as specified during the vibration

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(endurance) test and the reliability test. It is intended that all samples submitted for acceptance also meet this requirement. Any exceptions requested by the contractor to reduce production altimeter test or calibration time by making a zero adjustment after completion of acceptance tests shall establish that bias or calibration shifts did not occur during the acceptance tests and that the intended requirements are met. Adjustments may be permitted if the (average) difference in room temperature scale errors at the start of the group A, subgroup I inspections (for low altitude test points) do not differ from the scale errors recorded after completion of all group A, subgroup I inspections by more than ten feet. The re-zeroing procedures for such cases must be approved by the qualifying activity.

4.3.7 Standard pressure. The standard pressure used in calibrating and testing the altimeter shall be as specified in MIL-STD-859.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.4) on sample units that are manufactured by normal production procedures.

4.4.1 Sample. Three altimeters shall be subjected to the qualification inspection. The samples shall be taken at random from a production run. The sample altimeters shall have been subjected to and passed the requirements of group A inspection (see 4.5.1.2). Qualification shall not be granted if group A inspections are not met. Each altimeter type (I and II) shall be qualified separately.

4.4.2 Inspection routine. Sample altimeters shall be subjected to the qualification inspection specified in table VII in the order shown.

TABLE VII. Qualification inspection.

Inspection	Number of sample units	Requirement paragraph(s)	Method paragraph	Number of failures allowed
Examination of Product	3	3.2 through 3.3.4	4.6.1	1/
Manufacturing Run-in			4.6.2	
Barometric Setting Scale		3.4.3	4.6.3	
Barometric Setting Knob Functional Test		3.4.4	4.6.4	
Case Leakage		3.3.11.14	4.6.5	
Frictional Stop and Jump		3.3.11.18	4.6.6	
Static Friction		3.3.11.18	4.6.7	
Position Error		3.3.11.19	4.6.8	

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TABLE VII. Qualification inspection - Continued

Inspection	Number of sample units	Requirement paragraph(s)	Method paragraph	Number of failures allowed
Altitude Display Scale Error	3	3.3.11.17	4.6.9	1/
Hysteresis and After Effect		3.3.11.20	4.6.9.1	
Scale Error at -35°C		3.3.11.17	4.6.10	
Scale Error at +45°C		3.3.11.17	4.6.11	
Dielectric Strength		3.3.11.11	4.6.13	
Scale Error at -54°C		3.3.11.17	4.6.14	
Scale Error at +71°C		3.3.11.17	4.6.15	
Vibration Error		3.3.11.2	4.6.16	
Power Variation		3.3.13	4.6.17	
Response Test		3.3.11.10	4.6.19	
Baroset Knob, Torque, Pull and Push and Binding Test		3.4.4	4.6.20	
Shelf life		3.3.8	4.6.21	
Magnetic Property		3.3.11.12	4.6.22	
Electromagnetic Interference and Compatibility		3.3.5	4.6.23	
Overpressure Test		3.3.11.15	4.6.24	
Underpressure Test		3.3.11.16	4.6.25	
Pressurization Test		3.3.11.13	4.6.26	
Temperature-Altitude		3.3.11.1	4.6.27.1	
Vibration		3.3.11.2	4.6.27.2	
Acceleration		3.3.11.3	4.6.27.3	
Humidity		3.3.11.4	4.6.27.4	
Fungus		3.3.11.5	4.6.27.5	

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TABLE VII. Qualification inspection - Continued

Inspection	Number of sample units	Requirement paragraph(s)	Method paragraph	Number of failures allowed
Salt Spray	3	3.3.11.6	4.6.27.6	1/
Rain		3.3.11.7	4.6.27.7	
Sand and Dust		3.3.11.8	4.6.27.8	
Impact Shock		3.3.11.9	4.6.27.9	
Lighting		3.4.5	4.6.28	
Internal Examination		3.3.4	4.6.29	
Maintainability Demonstration		3.3.6	4.6.30	
Reliability Qualification Phase		3.3.2	4.6.31.1	

1/ No failures shall be permitted. Any discrepancies, failures, or unacceptable performance must be evaluated by the procurement/qualifying agency to determine if corrective action can be taken without repetition of any tests.

4.4.2.1 Qualification test sample identification. The qualification test samples shall be identified by durable tags, securely attached and marked with the following information:

Qualification Test Sample

Altimeter Type
Submitted by (manufacturer's name, date) For qualification test in accordance with MIL-A-85505(AS) under authorization (reference letter authorizing tests).

4.4.3 Failures. Failure to meet test requirements shall be cause for refusal to grant qualification.

4.4.4 Retention of qualification. To retain qualification, the contractor shall forward a report concerning test of production units at 2-year intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the tests performed for inspection of product for delivery (groups A and B), indicating as a minimum the number of altimeters that have passed, the number that have failed, and the group which they failed. The results of tests of all reworked altimeters shall be identified and accounted for.

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b. A summary of the results of tests performed for periodic inspection (group C), including the number and mode of failures. The summary shall include results of all periodic inspection tests performed and completed during the 2-year period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 2-year period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 2-year period that the inspection data indicates failure of the qualified product to meet the requirement of this specification. In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit his qualified products to testing in accordance with the qualification inspection requirements and the reason for no production.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A and B inspections.

4.5.1.1 Inspection lot. A normal lot size shall be 25 altimeters. Refer to table VIII. If the lot size is less than 25 altimeters, the quantity produced in two months shall be considered one lot. Unless otherwise noted by contract, concurrent delivery shall be permitted during the storage period. When an altimeter failure occurs, no items from those still on hand or later produced shall be accepted until the extent and cause of the failure has been determined and appropriately corrected. When a failure occurs, shift to one sample out of each 15 from those still on hand and proceed with sampling, as indicated in table VIII when the next quantity is offered for acceptance. In the event that a failure is noted for the sample which is subjected to the shelf life test only, the shelf life test is the sole test to which the increased sample must be subjected.

TABLE VIII. Group A inspection, subgroup II sample selection.

Quantity Offered for Acceptance	Quantity to be Selected for Tests
First 15	1
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

One additional sample, selected at random, shall also be tested from each monthly lot for shelf life only in accordance with 4.6.12.

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4.5.1.2 Group A inspection. Group A inspection shall consist of inspections specified in table IX. These inspections shall be performed in the order shown except as noted.

TABLE IX. Group A inspection.

Inspection	Requirement paragraph(s)	Test method paragraph	Sampling procedure
Subgroup I			
Examination of Products	3.2 through 3.3.4	4.6.1	
Manufacturing Run-in		4.6.2	
Barometric Setting Scale	3.4.3	4.6.3	
Barometric Setting Knob Functional Test	3.4.4	4.6.4	
Case Leakage	3.3.11.14	4.6.5	
Frictional Stop and Jump	3.3.11.18	4.6.6	
Static Friction	3.3.11.18	4.6.7	All sample units
Position Error	3.3.11.19	4.6.8	
Altitude Display Scale Error	3.3.11.17	4.6.9	
Hysteresis and After Effect	3.3.11.20	4.6.9.1	
Scale Error at -35°C	3.3.11.17	4.6.10	
Scale Error at +45°C	3.3.11.17	4.6.11	
Lighting Test	3.4.5	4.6.12	
Subgroup II			
Dielectric Strength	3.3.11.11	4.6.13	(See 4.5.1.2.1)
Scale Error at -54°C	3.3.11.17	4.6.14	
Scale Error at +71°C	3.3.11.17	4.6.15	
Vibration Error	3.3.11.2	4.6.16	
Power Variation	3.3.13	4.6.17	

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TABLE IX. Group A inspection - Continued

Inspection	Requirement paragraph(s)	Test method paragraph	Sampling procedure
Subgroup II Continued			
Lighting Tests	3.4.5	4.6.18	
Response Test	3.3.11.10	4.6.19	(See 4.5.1.2.1)
Baroset Knob, Torque, Pull and Push, and Binding Test	3.4.4	4.6.20	1 per sampling lot
Shelf Life	3.3.8	4.6.21	

4.5.1.2.1 Sampling plan. Subgroup I tests shall be performed on every altimeter supplied under this specification. Altimeters that do not meet the requirements of the subgroup I shall be removed from the lot. Lots having more than 20 percent total rejects shall not be furnished on contracts until the procuring activity is notified of the reject action and the corrective action being taken. Subgroup II samples shall be selected from production at random in accordance with the table VIII.

4.5.1.2.1.1 Manufacturer's production inspection. If the manufacturer performs tests similar to those specified in subgroup I, table IX as the final step of his production process, group A, subgroup I inspection may be waived and the data resulting from the manufacturer's production tests may be used instead. Authority to waive the subgroup I inspection shall be granted by the qualifying activity only. The following criteria must be complied with:

a. Tests conducted by the manufacturer during production shall be clearly identical to or more stringent than those specified for subgroup I. Test conditions shall be equal to or more stringent than those specified for subgroup I tests.

b. Manufacturer subjects 100 percent of the product supplied under this specification to his production tests.

c. The parameters measured and the failure criteria shall be the same or more stringent than those specified herein.

d. The lot rejection criteria is the same or more stringent than that specified herein.

e. The manufacturer shall make available all information concerning the test procedures and instrumentation used in the production tests. The manufacturer shall also make available to the Government all records of all detail test data resulting from production tests (see 6.2.2).

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f. Once approved the manufacturer shall not change the test procedures or criteria without prior notification and concurrence by the qualifying activity.

4.5.1.2.1.2 Rejected lots (subgroup II). Rejected inspection lots may be resubmitted for Government acceptance only if the manufacturer performs 100-percent inspection on altimeters of the lot for those characteristics which were defective and resulted in rejection of the lot, and removes all defective units and resubmits the lot for quality conformance inspection. Resubmitted lots shall be kept separate from new lots, and shall be clearly identified as resubmitted lots. Resubmitted lots shall be inspected using the tightened inspection procedure of MIL-STD-105 and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed.

4.5.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table X in the order shown. They shall be performed on sample units that have been subjected to and have passed the group A inspection.

4.5.1.3.1 Sampling plan. Three production samples shall be selected at random from the first 15 produced on contract and submitted 10 days after manufacture. Unless otherwise directed by contract or purchase order, these samples shall be forwarded at the contractor's expense to a Government laboratory designated by the procuring activity. Each sample shall be plainly identified by a securely attached, durable tag marked with the following information:

Altimeter Type
Submitted by (manufacturer's name, date) for production acceptance Group B inspection of MIL-A-85505(AS) with Contract No.
Manufacturer's part number

4.5.1.3.2 Rejected lots. If an inspection lot is rejected, the manufacturer may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.5.1.3.3 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered per the contract or purchase order, provided these units are not selected for group C inspection and meet requirements for visual and mechanical inspection.

4.5.2 Periodic inspection. Periodic inspection shall consist of group C. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.5.2.1.5), delivery of products which have passed groups A and B inspections shall not be delayed pending the results of these periodic inspections.

4.5.2.1 Group C inspection. Group C inspection shall consist of the tests specified in table XI, in the order shown. Group C inspection shall be performed on sample units of each style and selected from inspection lots which have passed groups A and B inspections. Group C inspection samples shall be representative of production.

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TABLE X. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	Numbers of failures allowed
Magnetic Property	3.3.11.12	4.6.22	2/
Electromagnetic Interference and Compatibility	3.3.5	4.6.23	
Overpressure Test	3.3.11.15	4.6.24	
Underpressure Test	3.3.11.16	4.6.25	
Pressurization Test	3.3.11.13	4.6.26	
Temperature	3.3.11.1	4.6.27.1	
Vibration	3.3.11.2	4.6.27.2	
Acceleration	3.3.11.3	4.6.27.3	
Humidity	3.3.11.4	4.6.27.4	
Fungus	3.3.11.5	4.6.27.5	
Salt Spray	3.3.11.6	4.6.27.6	
Rain	3.3.11.7	4.6.27.7	
Sand and Dust	3.3.11.8	4.6.27.8	
Impact Shock	3.3.11.9	4.6.27.9	
Lighting	3.4.5	4.6.28	
1/ Internal Examination	3.3.1, 3.3.4	4.6.29	
Maintainability Demonstration	3.3.6	4.6.30	

1/ The altimeter selected for internal examination shall not be exposed to the fungus test.

2/ No failures shall be permitted. Any discrepancies, failures, or unacceptable performance must be evaluated by the procurement/qualifying agency to determine if corrective action can be taken without repetition of any tests.

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TABLE XI. Group C inspection.

Test or Inspection	Requirement Paragraph	Test Method Paragraph	Number of Sample Units for Inspection
Reliability Sampling Phase Tests	3.3.2.3	4.6.31.2	3-6 units per month
Longevity Tests	3.3.2.2	4.6.31.3	As required

4.5.2.1.1 Sampling plan. Three (3) to six (6) altimeters shall be selected at random from the monthly lot for acceptance-reject criteria of test plan IV of MIL-STD-781. If the lot size is less than 25 altimeters, the quantity produced in two months shall be made available for the procuring activity. At any time that the test results show a reject situation, the procuring activity shall be notified (see 6.2.2).

4.5.2.1.2 Failure criteria. If recalibration, adjustment, or repair is required to pass any of the tests listed in 4.6.31.4, either at the 250 hour test intervals or at the completion of the reliability test, a failure shall be charged to that test. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure unless one part caused the other parts to fail. A 25 foot out-of-tolerance condition in scale error will be permitted after each 250 hour cycling period, provided rezeroing by used of the barometric setting system will correct the condition. An increase in friction tolerances to the values indicated as "final" in table III is permitted upon completion of 1000 hours of reliability cycling. Any malfunction of the internal vibrator shall be charged as a failure. (The 25 foot out-of-tolerance allowed for the scale after each 250 hour cycling period shall be applied as a data correction and shall not be a physical correction by means of the zero setting system.)

4.5.2.1.3 Preventive maintenance. No preventive maintenance shall be performed on the altimeter during the 1,000 hour MTBF test. The failures occuring during the longevity test shall be analyzed in accordance with MIL-STD-781.

4.5.2.1.4 Disposition of samples. Any group C sample tested may be delivered on contract provided the sample is in "good as new" condition or has been refurbished and meets the group A, subgroup I inspections and is representative of production units currently being accepted.

4.5.2.1.5 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure (see 6.2.2) and take corrective action on the materials or processes, or both, as warranted, and on all units or product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance

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and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be made available for the cognizant inspection activity and the qualifying activity (see 6.2.2).

4.5.3 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 5.

4.6 Methods of inspection. The procedures used for conducting qualification, and quality conformance tests shall be prepared by the contractor in accordance with MIL-T-18303 and the following paragraphs for procuring activity review and approval (see 6.2.2). The right is reserved by the procuring activity to modify the test or require additional tests deemed necessary to determine compliance with the requirements of this specification or the contract.

4.6.1 Examination of product.

a. Qualification tests. Prior to final assembly of the altimeter to be subjected to qualification tests, a detailed examination shall be made to verify compliance with each of the design requirements of section 3 that are not covered by specific tests.

b. Quality conformance tests. Each altimeter submitted for acceptance shall be examined externally to determine comformance with the applicable drawings and with each of the requirements of this specification capable of being inspected in this manner.

4.6.2 Manufacturing run-in. Each altimeter shall be subjected to a run-in test of six hours duration. The first two hours of the run-in test shall be accomplished with the altimeter exposed to +71°C temperature. The remaining four hours of the test shall be accomplished under room ambient conditions with the altimeter mounted on a special test table. The table shall subject the altimeter to roll, pitch, and yaw motion (Scorsby) at a frequency of five to seven oscillations per minute through an angle of 30 degrees. Prior to initiation of the run-in test, the scale error at zero altitude 29.92 in. Hg. and the electrical output of the barometric setting shall be determined and recorded. The scale error shall be within the room temperature scale error tolerance of table II, and the electrical output shall be within the tolerances of table V. During the performance of the entire run-in tests, the altimeter shall be subjected to vibration of approximately 1/2 g at a frequencies between 20 and 60 Hz. External vibration need not be applied if the specified 1/2 g or greater vibration level is achieved by use of the integral vibrators only. The 5V lighting circuit and the barometric output potentiometers shall be energized at all times during the test. The power to operate the integral vibrators (28 VDC) shall also be applied continuously during the test performed at high temperature and at ambient room temperature. The altimeter shall be pneumatically cycled during the six hour run-in test. The pneumatic cycle shall range from approximately sea level pressure to about 35,000 feet altitude and return to sea level within approximately 15 minutes for a total of at least 4 cycles per hour. The barometric setting counters shall not have changed by more than 0.01 in. Hg. from the inital setting after completion of the run-in test. Failures noted for production altimeters during the run-in test shall be summarized and added to the monthly production reliability test report prepared by the contractor.

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4.6.3 Barometric setting scale. The altimeter shall be subjected to atmospheric pressure. The pressure scale shall be set successively to the values specified in table V for both the clockwise and counter-clockwise directions of rotation of the baroset knob. The indicated altitude and barometric potentiometer output at each test point shall be recorded. After the series of readings have been recorded, the difference between the altimeter indicator at the setting of 29.92 and each other setting shall be found. (The reading obtained at the 29.92 setting for the clockwise direction of rotation of the baroset knob shall be used to compute the differences for both the clockwise and counter-clockwise directions of rotations.) The differences shall be recorded and compared with the values in the "Correct Difference" column of table V. The recorded differences shall not differ from the correct differences by more than ±25 feet at any setting. The difference between the value recorded for the clockwise and counter-clockwise rotation of the baroset knob shall not exceed 20 feet. The data obtained for the potentiometer output shall also be processed in the same manner and the results shall be within the tolerances of table V. Atmospheric pressure shall be recorded before and after this test and proper allowance shall be made for any variation. (The design of the altimeter shall be such that the test requirement shall be met under any atmospheric pressure within the range -1,000 to +6,000 feet.) (This test may be performed with the static port connected to a controlled pressure source if provisions are made to ensure that some altimeters of each production lot are subjected to controlled pressures at settings other than 29.92.)

4.6.4 Barosetting knob functional test. The barometric setting system shall be operated slowly from the lowest setting possible to the highest pressure setting and return. The torque required for operation from stop to stop throughout this range shall be measured and shall be 2-7 inch ounces. The electrical output of barometric setting shall be observed to meet the requirements of 3.4.1 or 3.4.2, as applicable, when the stops are approached.

4.6.5 Case Leakage. A vacuum sufficient to produce a reading of 3,000 feet (approximately) above ambient altitude shall be applied to the static pressure port of the altimeter, at which point the connecting tubing shall be pinched off or otherwise completely sealed. The change in reading in one minute shall not exceed 100 feet. The leakage shall be checked in this manner for each 90° position of the baroset knob. The leakage check for each 90° position of the baroset knob is applicable only to room temperature tests. This test shall be repeated for an altimeter reading of 40,000 feet altitude (approximately). Altimeters shall be tested individually. This test may be combined with Altitude Display Scale Error Test (see 4.6.9).

4.6.6 Friction stop and jump. The altimeter shall be tested for friction in both ascending and descending directions at the test points specified in table III. With the internal vibrator energized, the pressure to the static port of the altimeter shall be applied at the rates indicated for the test points in table III. The change in the indicating position of the pointer, due to apparent stopping and jumping as the counter is turning or at any other place around the dial, shall be recorded as friction and shall not exceed the tolerances specified in table III, column C. This test shall be repeated with power to the integral vibrator off. The stop and jump friction for the test as specified shall not exceed the tolerances specified in table III, column D. After the preceding requirements are met, the altimeter shall again be subjected to the vibrator operative test, except the rate-of-change of altitude shall be 20,000 feet per minute (fpm) and the face of the altimeter shall be inclined down 45° during descent and inclined up 45° during ascent.

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The tolerance during this portion of the test shall be increased by 100 feet over the values listed in column C of table III. The barometric pressure setting may be set at any point within its range during this test. (When the data from a production lot indicates that the portion of the test performed for 45° inclination angles and at 20,000 feet per minute altitude change rates is satisfactory for every altimeter of the lot, the contractor may perform this test on a sampling plan basis only. In the event of failure to meet the requirements for this portion of the test during group A or B inspections, the contractor shall immediately revert to 100% tests of all production units.)

4.6.7 Static friction. The altimeter shall be subjected to this test with all electrical power off. A decreasing pressure not exceeding 3000 fpm shall be applied to the static pressure port. As the static friction test point listed in table IV is approached, the rate of pressure change shall be gradually reduced until the desired test point is attained. The pressure shall then be held constant while two readings of the altimeter are taken, the first without the integral vibrator operating and the second with the vibrator operating. The difference between the two readings shall be recorded as static friction and shall be not greater than the tolerances listed in table IV.

4.6.8 Position error. The altimeter position error display readings shall be taken while the altimeter is vented to atmosphere and in each of the following positions:

a. Normal operating position

b. Rotated about the longitudinal axis 90°

c. Rotated about the longitudinal axis 180°

d. Rotated about the longitudinal axis 270°

e. Dial face up

The reading in any test position shall not differ from the reading recorded in any other test position listed by more than 20 feet. This test shall be performed with 28 VDC vibrator power applied to the altimeter. The test procedure shall permit tests of altimeters with the pointer located at any point around the dial.

4.6.9 Altitude display scale error. For a period of not less than 4 hours prior to this test, the altimeter shall not have been subjected to any testing involving operation at other than atmospheric pressure. The altimeter shall be tested for scale errors as follows: The barometric setting shall be set at 29.92 in. Hg. and shall remain at this setting during performance of all scale error tests. The direction at which the 29.92 in. Hg. setting point is approached shall not be a factor in determining scale error accuracy. The resultant scale errors shall be within the specified tolerance regardless of the direction of approaching the setting. The altimeter shall then be subjected throughout the applicable range to successively decreasing pressure and then increasing pressure at the test points specified in table XII. The altimeter shall remain at the pressure corresponding to each test point for at least one minute before a test reading is taken. The scale errors shall not exceed the tolerances specified in table XII. This test may be performed

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with the altimeter in an altitude chamber. During this test, the movement of the altimeter counter pointer display shall be free from apparent backlash and any irregular motion when the pressure is varied uniformly.

4.6.9.1 Hysteresis and after effect. Tests of this paragraph may be combined with the Altitude Display Scale Error test (see 4.6.9). Not more than 15 minutes after the altimeter has been first subjected to the pressure corresponding to 50,000 feet, the pressure shall be increased at a rate corresponding to a decrease in altitude of approximately 20,000 fpm until the pressure corresponding to the test point given in table XII is reached. The altimeter shall remain at this pressure for at least 1 but not more than 15 minutes before the test reading is taken. The pressure shall again be increased at the same rate and the procedure repeated for the 20,000 feet and 0 feet test points. The readings of the altimeter at the three test points shall not differ from the readings of the instrument for the same test points in the prior scale error by tests more than the tolerances specified in table XII. In addition, all test readings shall be within the scale error tolerances of table II.

TABLE XII. Hysteresis.

Inches Hg. Test Point	Standard Altitude	Permissible Variations
1st 11.103	25,000 feet	70 feet
2nd 13.750	20,000 feet	70 feet
3rd 29.921	0 feet	40 feet

4.6.10 Scale error at -35°C. The altimeter without electrical power applied shall be subjected to a temperature of -35°C for a period of three hours. At the end of this time period, all electrical power shall be applied to the altimeter, and the Case Leakage (see 4.6.5) and Altitude Display Scale Error lists (see 4.6.9) shall be performed while maintaining temperature of -35°C. The integral lights shall be observed to be on. The potentiometer output of the altimeter shall also be recorded for the 29.92 in. Hg. setting only. The test points and tolerances shall be as specified in tables II and V. At the completion of this test and after the altimeter has stabilized at room temperature, the scale error and potentiometer output at zero altitude test point only shall be determined. The room temperature tolerances shall apply for this check. During production testing, the scale error and potentiometer output at zero altitude checks need not be performed if the scale error at 45°C test per 4.6.11 is performed subsequent to the test at -35°C.

4.6.11 Scale error at 45°C. The tests specified in 4.6.10 (Scale Error at -35°C) shall be repeated except the temperature shall be +45°C. At the completion of this test and after the altimeter has stabilized at room temperature, the scale error and the potentiometer output at the zero altitude test point only shall be determined. The room temperature tolerances shall apply for this check. During production testing, the scale error at zero altitude and potentiometer output checks need not be performed if the scale error at -35°C test in accordance with 4.6.10 is performed and subsequent to the test at +45°C.

4.6.12 Lighting (Group A inspection, subgroup I). The lighting system shall be excited with 5.0 +0.1 V and examined to verify that all lamps function properly.

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4.6.13 Dielectric strength. The altimeter shall be checked for insulation resistance by testing from the terminals indicated to ground at the potentials indicated for no more than 30 seconds. (The maximum voltage shall be raised from zero by rheostat control. Switching to the voltage from zero shall not be permitted.) There shall be no breakdown of insulation or any other permanent damage to the indicator as a result of this test.

4.6.14 Scale error at -54°C. The test specified in 4.6.8 (Altitude Display Scale Error) shall be repeated except that during the test and for a period of 3 hours without power applied before the test, the altimeter shall be at a temperature of -54° ±2°C. The test shall be initiated within 2 minutes after power is applied. The test points and tolerances shall be as specified in table II. The potentiometer output should be recorded for the 29.92 in. Hg. barometric pressure setting and should meet the tolerances specified in table V. At the completion of this test and after the altimeter has stabilized at room temperature, the scale error and the barometric potentiometer output at the zero altitude test point only shall be determined. The room temperature scale error tolerance shall apply for this check. During production testing, the zero altitude checks need not be performed if the scale error at +71°C is performed subsequent to the test at -54°C.

4.6.15 Scale error at +71°C. The test specified in 4.6.14 (Scale Error at -54°C) shall be repeated except the test temperature shall be +71 ±2°C. At the completion of this test and after the altimeter has stabilized at room temperature, the scale error and the barometric setting potentiometric error at the zero altitude test point only shall be determined. The room temperature scale error tolerance shall apply for this check. During production testing, the zero altitude checks need not be performed if the scale error at -54°C is performed subsequent to the test at +71°C.

4.6.16. Vibration error. The altimeter shall be mounted on the vibration stand in its normal operating position and secured by flange screws in the same position as on an aircraft panel. The barometric pressure setting and potentiometer output shall be recorded. The altimeter shall be subjected to one hour of vibration with an amplitude between 0.003 and 0.005 inch at frequency which is varied uniformly from 500 CPM to 3,000 CPM and return to 500 CPM during the hour. The vibration shall be such that any point of the altimeter case will describe a circular path of the amplitude stated, inclined at any angle of 45°C to the horizontal and dial planes. The natural frequency of the altimeter shall not occur in this frequency range. While the altimeter is being vibrated, the maximum amplitude of the pointer oscillation shall not exceed 20 feet and the pointer variation from normal shall not exceed 20 feet. Also, while under vibration, the barometric setting potentiometer shall not have intermittent output. No shift in the barometric setting counter indication shall occur with respect to the setting recorded prior to vibration. The integral lighting of the altimeter shall be operated during this test and shall not flicker. Upon completion of the preceding vibration cycling, the one hour test shall be repeated except that the amplitude of vibration shall be set between 0.009 and 0.011 inch. Under this test condition, the maximum amplitude of pointer oscillation shall not exceed 30 feet and the pointer variation from the normal shall not exceed 30 feet. The potentiometer output of the barometric pressure setting shall be monitored for continuity and accuracy and shall remain within the tolerances of table V. No shift in the barometric scale counter shall occur with respect to the settings recorded prior to vibration. A change in the barometric setting by 1/2 digit (0.005 in. Hg.) during or after the performance of the vibration test is

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defined as a shift. The integral lighting of the altimeter shall be operated during this test and shall not flicker. At the completion of the vibration error test, the scale error at zero altitude test point shall be determined. The room temperature scale error tolerance shall apply for this check. All the post shelf life tests listed in 4.6.21, must be performed after completion of the vibration test if for any reason all test samples will not be subjected to the shelf life tests.

4.6.16.1 Alternate vibration error. An equivalent vibration test using a linear shaker may be performed as an alternative to the preceding test. The altimeter shall be vibrated in three mutually perpendicular directions at the appropriate amplitudes. The procedures for the test shall be in accordance with MIL-STD-810 and shall be approved by the procuring and qualifying activity.

4.6.17 Power variation. The altimeter shall be tested for effects of power variation on performance. The vibrator shall be energized with 28 volts dc applied to the appropriate electrical pins. The vibrator excitation voltage shall then be reduced to 20 volts and the altimeter subjected to the scale error test and the friction stop and jump test. The indicated test points of tables II, III, and IV shall be used and the tolerances of tables II, III, and IV, column C shall apply. The vibrator excitation voltage shall then be reduced to 16 volts for a period of one minute. There shall be no damage to the vibrator. When the excitation voltage is returned to normal, the vibrator shall operate satisfactorily. The polarity of the vibrator excitation voltage shall then be reversed for a period of one minute. There shall be no damage to the vibrator or associated electrical circuitry as a result of this test.

4.6.18 Lighting tests (Group A inspection, subgroup II). Lighting tests specified as Sampling Plan A tests in MIL-L-25467 and MIL-L-27160 shall be conducted. Those tests which are common may be conducted once.

4.6.19 Response test. The altimeter shall be subjected to an increasing and decreasing rate of change of pressure equivalent to 30,000 feet/minute and operated throughout its range. No damage to the internal mechanism shall occur due to this test, and no zero or calibration shift shall result. Operation of the counter pointer display shall operate freely during the test. The altimeter shall then be subjected to rates of change of pressure equivalent to 50,000 feet/minute for type I altimeters or 80,000 feet/minute for type II altimeters as applicable without damage to the internal mechanism. At the completion of this test, the altimeter shall be checked for scale error at the zero test point.

4.6.20 Baroset knob torque, push, pull and bending test. The following tests shall be performed to demonstrate that the design meets the requirements specified in 3.4.4 (Barometric Setting System).

a. The setting range and presence of the required stops shall be verified. A twisting force of 6 pound inches maximum shall be applied at the barosetting knob against each stop. The stops shall be verified to be located as required by 3.4.4 and the mechanism shall not be damaged or the calibration affected by the torque applied. When a mechanism is incorporated in the altimeter to limit the torque applied to the stops, the torque limit prior to slip shall be recorded and satisfactory operation of the mechanism demonstrated. The travel of the altitude pointer for one turn of the barometric

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setting knob shall be within the range of 200 to 300 feet. Clockwise rotation of the setting knob shall result in clockwise rotation of the pointer.

b. A pulling force of 15 pounds shall be applied to the barosetting knob. While the force is applied, the knob shall be rotated at least 90°. This action shall be repeated with a pushing force of 15 pounds applied. The relationship of the potentiometer output and barometric setting indication shall not be affected by the forces applied and the mechanism shall not be otherwise damaged or the scale error affected.

c. The barosetting knob shall withstand a force of 15 pounds applied at the extreme outward end of the knob without permanent deflection or damaging of the mechanism.

4.6.21 Shelf life. The altimeter shall be placed in storage for a period of not less than 30 days. During the 30 day period, the altimeter shall remain as dormant as practical. The mechanism shall not be exercised other than that caused by normal atmospheric pressure variation. The temperature shall remain constant; however, normal room temperature variations will be permitted. At the end of the 30 day period, the altimeter shall be subjected to and meet the requirements of the tests listed in table XIII.

TABLE XIII. Post shelf life test.

Test	Requirement paragraph
Barometric Setting Scale 1/	4.6.3
Case Leakage	4.6.5
Friction Stop and Jump	4.6.6
Static Friction	4.6.7
Position Error	4.6.8
Altitude Display Scale Error	4.6.9
Hysteresis and After Effect	4.6.9.1
Lighting (Group A inspection, Subgroup I)	4.6.12

1/ Rezeroing of the barometric setting mechanism shall not be required to meet scale error tolerances.

4.6.22 Magnetic property. The altimeter, not energized, shall be rotated in a vertical plane about a short-bar magnetic compass with the nearest part of the altimeter 5 1/2 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the altimeter shall be held in position 0°,

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45°, 90°, 135°, 180°, 225°, 270° and 315° from the initial position. At each of these positions, the altimeter shall be rotated about on its own horizontal axis until it is in the normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. With the altimeter at any specified position, the compass defection shall be not greater than 3°. This test shall be repeated with the altimeter energized with rated voltage.

4.6.23 Electromagnetic interference and compatibility (EMIC). The altimeter shall be subjected to the following EMIC tests in accordance with the requirements of MIL-STD-461, class A1b and appropriate test methods of MIL-STD-462. During the tests, the altimeter shall be electrically energized and the potentiometer output shall be connected to a circuit simulating an aircraft installation. Provision shall be made to turn the barometric setting knob in order to vary the potentiometer output during the tests. During the susceptibility tests, the test pressure to the altimeter shall be varied and the performance of the integral vibrator shall be observed with respect to removing friction. Any effect on altimeter accuracy with the pressure held constant shall be noted.

4.6.23.1 Conducted emissions, power and interconnecting leads, 0.15 to 50 MHz. Conducted emissions measurements shall be made over the frequency range of 0.15 to 50 MHz on power and interconnecting leads in accordance with appropriate test methods of MIL-STD-462.

4.6.23.2 Conducted emissions, power leads, spikes, time domain. Conducted emissions measurements shall be made for spikes (switching transients) on power leads in accordance with appropriate test methods of MIL-STD-462.

4.6.23.3 Conducted susceptibility, power leads, 30 Hz to 400 MHz. The altimeter shall be subjected to conducted susceptibility tests by having signals over the range of 30 Hz to 400 MHz injected on the power leads in accordance with appropriate test methods of MIL-STD-462. Evidence of susceptibility observed shall be recorded (see 4.6.23).

4.6.23.4 Conducted susceptibility, spikes, power leads. The altimeter shall be subjected to conducted susceptibility tests consisting of superimposing a spike on the power line voltage waveform in accordance with appropriate test methods of MIL-STD-462. Evidence of susceptibility observed shall be recorded (see 4.6.23).

4.6.23.5 Radiated emissions, electric field, 14 kHz to 10 GHz. Radiated emissions emanating from the altimeter shall be measured over the frequency range of 14 kHz to 10 GHz in accordance with appropriate test methods of MIL-STD-462.

4.6.23.6 Radiated susceptibility, magnetic field, 0.03 to 50 kHz. The altimeter shall be subjected to radiated susceptibility tests over the frequency range of 0.03 to 50 kHz, magnetic field, in accordance with appropriate test methods of MIL-STD-462. Evidence of susceptibility observed shall be recorded (see 4.6.23).

4.6.23.7 Radiated susceptibility, magnetic induction field, spikes, and power frequencies. The altimeter shall be subjected to radiated susceptibility, magnetic inducting spikes, and power frequencies (part I and II) in accordance with appropriate test methods of MIL-STD-462. Evidence of susceptibility observed shall be recorded (see 4.6.23).

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4.6.24 Overpressure test. With the barometric setting set at 29.92, a pressure of 35 in. Hg. absolute shall be applied to the static port of the altimeter for a period of one minute. No damage to the internal mechanism shall occur due to this pressure and no zero or calibration shift shall result. At the completion of this test, the altimeter shall be checked for scale error at the zero altitude.

4.6.25 Underpressure test. Type I altimeters shall be subjected to an altitude of 60,000 feet for a period of one minute. Type II altimeters shall be subjected to an altitude of 100,000 feet for a period of one minute. No damage to the internal mechanism shall occur as a result of exposure to this pressure and no zero or calibration shift shall result. At the completion of this test, the altimeter shall be checked for scale error at the zero altitude test point.

4.6.26 Pressurization test. The altimeter shall be placed in a pressure chamber with the pressure port connected to outside ambient pressure (29.92 inches of mercury approximately) during this test. The chamber pressure shall be raised to 60 in. Hg. absolute and maintained at approximately 25°C for a period of 4 hours. During the 4 hour period, no deformation or other damage to the altimeter shall occur. At the completion of the test, the altimeter shall be checked for scale error at the zero altitude test point.

4.6.27 Environmental tests. The altimeter shall be subjected to the following tests to be performed in accordance with the specified procedures of MIL-STD-810. Upon completion of each environmental test, the altimeter shall be subjected to and meet the tests of 4.6.5 (Case Leak), 4.6.3 (Barometric Setting Scale), 4.6.6 (Friction Stop and Jump), 4.6.7 (Static Friction), 4.6.9 (Altitude Display Scale Error), and 4.6.12 (Lighting Group A Inspection, Subgroup I). The indicated test points of tables II, III, IV, and V shall apply. When external connections are required to simulate installed conditions, the connections shall consist of standard AN or MS fittings and at least 10 feet of 1/4 or 3/8 inch tubing, coiled and so arranged as to allow drainage of condensed vapors. The tubing shall be the static pressure port of the altimeter.

4.6.27.1 Temperature.

4.6.27.1.1 Low temperature exposure. The low temperature exposure test shall be performed in accordance with MIL-STD-810, Method 502.1, Procedure 1. The operating temperature shall be -54°C. The operational tests at -54°C shall consist of an altitude display scale error test and a verification of satisfactory barometric pressure setting output for the 29.92 inch Hg. setting.

4.6.27.1.2 High temperature exposure. The high temperature exposure test shall be performed in accordance with MIL-STD-810, Method 501.1, Procedure 1. The exposure temperature shall be 95°C. The period of exposure shall be 4 hours. The temperature at which operational tests are to be performed shall be 71°C. The operational tests at 71°C shall consist of an altitude display scale error test and a verification of satisfactory barometric pressure setting output for the 29.92 inch Hg. setting.

4.6.27.2 Vibration.

4.6.27.2.1 General. The altimeter shall be subjected to vibration tests in accordance with MIL-STD-810 and tables XIV and XV of this specification. The altime-

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TABLE XIV. Vibration test (Non-gunfire).

Frequency Range (Hz)	Type of Motion	Test Time Per Axis		MIL-STD-810 Applicable Curve	Altimeter Test Requirement
		Sweep Time	Sweep Rate Cycle		
5-50	Sinusoidal Cycling	30 minutes (Includes resonance dwell of 5 minutes)	7.5 minutes	Figure 514.2-2, 5-33 Hz, Curve C; 33-50 Hz, Curve B (2g)	Functional (Performance)
50-2000	Random	30 minutes	20 minutes	Figure 514.2-2A, 50-1000 Hz; Maximum acceleration power spectral density Wo = 0.009 G^2/Hz. Minimum overall g rms = 3.6. 1000-2000 Hz; -6dB/octave	Functional (Performance)
5-50	Sinusoidal Cycling	90 minutes (Includes resonance dwell time 30 minutes)	7.5 minutes	Figure 514.2-2, 5-33 Hz, Curve C; 33-50 Hz, Curve B (2g)	Endurance (Non-functional)
50-2000	Random	60 minutes	20 minutes	Figure 514.2-2A, 50-1000 Hz; Maximum acceleration power spectral density Wo = 0.027 G^2/Hz. Minimum overall g rms = 6.3 1000-2000 Hz; -6dB/octave	Endurance (Non-functional)

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TABLE XV. Vibration test (Gunfire)

Frequency Range (Hz)	Type of Motion	Test Time Per Axis		MIL-STD-810 Applicable Curve	Altimeter Test Requirement
		Sweep Time	Sweep Rate Cycle		
50-500	Sinusoidal Cycling	60 minutes	7.5 minutes	Figure 514.2-2, 50-60 Hz; 0.025 inch double amplitude (D.A.) to ±3.5 g 60-220 Hz; ±3.5 g to 0.00156 D.A. 220-500 Hz; 0.00156 to ±7 g	Endurance (Non-functional)
500-2000	Random	15 minutes		Figure 514.2-2A, 500-1000 Hz; Maximum acceleration power spectral density Wo = 0.067 G^2/Hz; Minimum overall g rms 7.6 1000-2000 Hz, -9dB/octave	Endurance (Non-functional)

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ter shall be mounted in the normal operating position for all vibration tests and shall be electrically energized and operating.

4.6.27.2.2 Vibration resonance search. During vibration resonance searches as required to be performed as part of sinusoidal cycling functional tests, the pressure to the altimeter shall be maintained constant or at room ambient. As the resonant search is performed, the altimeter display (pointer) shall not exhibit oscillations greater than 20 feet and pointer variation from the normal shall not exceed 20 feet.

4.6.27.2.3 Post resonant search sinusoidal cycling vibration test. Upon completion of the resonance search, the remainder of the sinusoidal cycling functional vibration test for each axis shall be performed while the altimeter is being pressure cycled. The pressure to the altimeter shall be cycled from room ambient to approximately 30,000 to 40,000 feet altitude and return to ambient at a rate of about 2000-4000 feet per minute. The function of the altimeter under test shall be noted to be satisfactory with respect to smoothness of operation (friction, stop, and jump).

4.6.27.2.4 Random vibration test. The altimeter shall be pressure cycled during random vibration functional test and performance (friction, stop and jump) shall be satisfactory during the cycling, but a formal test shall not be required. After the completion of functional vibration tests, the altimeter barometric pressure setting shall be noted to have drifted no more than 0.01 inch Hg. with respect to the setting recorded prior to the tests. The altimeter shall also be subjected to, and meet the requirements of the individual tests listed in 4.6.27.

4.6.27.2.5 Vibration endurance test. During vibration endurance tests (non-gunfire and gunfire), the altimeter shall be energized and pressure cycled, but functional performance effects on the altimeter need not be evaluated. After the completion of functional tests, the altimeter barometric pressure setting shall be noted to have drifted no more than 0.01 in. Hg. with respect to the setting recorded prior to the tests.

4.6.27.2.6 Post vibration endurance test. After completion of all the endurance tests, the altimeter shall again be subjected to the tests listed in 4.6.27. The results of these tests shall be within tolerances specified. The tolerances identified as "Final" in tables III and IV shall apply to static friction and stop and jump friction tests. A scale error (bias) shift of 50 feet shall also be permitted if the scale error requirements of table II can be met by means of the barometric pressure zero setting adjustment, 3.4.4.2. During vibration endurance tests, the barometric setting counter shall not change by more than 0.01 in. Hg. with respect to the setting recorded prior to the test.

4.6.27.3 Acceleration. Acceleration tests shall be performed in accordance with MIL-STD-810, method 513.2, procedure I for structural test and procedure II for operational test.

4.6.27.3.1 Procedure II, operational test. The altimeter shall be subjected to tests on a centrifuge while oriented in each of the six positions defined by MIL-STD-810, method 513.2. The altimeter shall be installed so that it can be electrically energized and pneumatically operated during the test. The altimeter shall be subjected to the "g" test of table XVI.

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TABLE XVI. Operational test "g" level.

Altimeter Position (Orientation)	Centrifuge "g" Test Level	
	Type I Altimeter	Type II Altimeter
Fore	1.0	2.0
Aft	1.0	6.0
Up	7.6	9.0
Down	5.3	3.0
Lateral, Right Side	2.2	4.0
Lateral, Left Side	2.2	4.0

The test shall be performed with the altimeter static pressure port exposed to ambient pressure and shall be repeated with the altimeter exposed to the altitude pressure of approximately 25,000 feet (11.103 in. Hg.). The change in altimeter scale errors recorded for each position during acceleration shall not exceed 20 feet per "g". The electrical output of barometric setting shall be within the tolerances specified by 4.6.3 for the 29.92 in. Hg. test point. The altimeter lighting shall be noted to be continuous (without flicker) during acceleration. The tests listed in 4.6.27 shall be performed upon completion of this procedure.

4.6.27.3.2 Procedure I, structural test. The altimeter shall be subjected to tests on a centrifuge while oriented in each of the six positions defined by MIL-STD-810, method 513.2. The altimeter shall not be electrically energized or pneumatically operated during this test. The altimeter shall be subjected to the "g" level test of table XVII. Upon completion of this procedure, the altimeter shall be subjected to and shall meet the requirements of the tests listed in 4.6.27.

4.6.27.4 Humidity. The humidity test shall be performed in accordance with MIL-STD-810, method 507.1, procedure I, except the test item may be removed from the chamber at the completion of the exposure period for the performance of tests. The altimeter shall be thoroughly examined within at least 48 hours after completion of the humidity exposure and there shall be no evidence of corrosion and deterioration that would affect subsequent operation of the altimeter.

4.6.27.5 Fungus. The fungus test shall be performed in accordance with MIL-STD-810, method 508.2. At the end of test period, the altimeter shall be thoroughly examined to ascertain that no fungus growth has occurred. This test need not be performed if a detailed review of all altimeter components is made and appropriate certification is provided concerning the use of only non-nutrient material in accordance with MIL-STD-454, requirement 4.

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TABLE XVII. Structural test "g" level.

Altimeter Position (Orientation)	Centrifuge "g" test level
Fore	3.0
Aft	9.0
Up	13.5
Down	4.5
Lateral (Right Side)	6.0
Lateral (Left Side)	6.0

4.6.27.6 Salt spray. The salt spray test shall be performed in accordance with MIL-STD-810, method 509.1, procedure I. The altimeter shall be subjected to the test listed in 4.6.27 (Environmental Tests) immediately after completion of the salt spray exposure period and again 48 hours (approximately) later.

4.6.27.7 Rain. The rain test shall be performed in accordance with MIL-STD-810, method 506.1, procedure I. Operation of the altimeter during exposure to the rain shall not be required. The front surface (dial) only shall be directly exposed to the rain. The test duration shall be for a period of 30 minutes.

4.6.27.8 Sand and dust. The sand and dust test shall be performed in accordance with MIL-STD-810, method 510.1, procedure I.

4.6.27.9 Shock.

4.6.27.9.1 Basic design shock test. The basic design shock test shall be performed in accordance with MIL-STD-810, method 516.2, procedure I, figure 516.2-1, peak value (a), time duration (c). The altimeter shall be electrically energized and operating during the test and the static pressure port shall be open to ambient pressure. Prior to the application of impacts, the altitude indication and the barometric setting potentiometer output shall be recorded for a setting of 29.92 in. Hg. Monitoring altimeter performance during application of impact shocks is not required. Upon completion of the tests for each axis, the altitude indication, barometric setting, and the potentiometer output shall again be recorded and this data compared to data taken prior to the application of the impacts. The altimeter and data shall be examined, and any electrical or mechanical failure or degradation shall be recorded. Upon completion of the basic design shock test, the altimeter shall be subjected to and meet the requirements of the individual tests listed in 4.6.27.

4.6.27.9.2 Crash safety shock test. This test shall be performed in accordance with MIL-STD-810, method 516.2, procedure III, figure 516.2-1, peak value (a), time duration (c). A dummy load may be used in lieu of an altimeter. Upon completion of

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the crash safety shock test, the test item shall be examined and all damage recorded. Bending and distortion of the mounting attachment shall not constitute a failure, but a broken mounting attachment shall be counted as an equipment failure. Operation of the indicator after the shock test is not required.

4.6.28 Lighting (Group C inspection subgroup 1). Lighting tests shall be conducted in accordance with MIL-L-25467 and MIL-L-27160, sampling plan B tests. These tests which are common to both specifications may be conducted once. Integral lighting colors shall be in accordance with MIL-C-25050.

4.6.29 Internal examination. Only one altimeter shall be subjected to this test. The barometric setting locking screw of the altimeter shall be loosened and examined to determine if the barometric setting system meets the requirements of 3.4.4 and 3.4.4.2. The range of adjustment of the barometric scale counter relative to the altimeter pointer shall be recorded. The range of adjustment of the potentiometer output relative to the barometric counter shall also be recorded if such an adjustment is provided. The case of the altimeter shall then be opened and the internal mechanism shall be examined to determine conformance with the applicable requirements of this specification. The lamp circuit of the altimeter shall also be examined for conformance to the requirements of MIL-L-25467. In addition, any damage or deterioration to the mechanism that resulted from any of the preceding tests, or any other deficiency which could in any manner prevent the altimeter from meeting operational or maintenance requirements during service life, shall be cause for rejection. The qualification sample altimeter shall be re-cased and demonstrated to perform satisfactorily using abbreviated room temperature tests. Altimeter shall be cleaned prior to re-casing.

4.6.30 Maintainability demonstration. Maintainability demonstration shall be in accordance with MIL-STD-471, test method 1, plan B1. Maintainability demonstration shall be performed on a minimum of three selected repairs. The tasks selected to demonstrate maintainability shall be selected in advance and shall be approved by the procuring activity (see 6.2.1). An equivalent maintainability demonstration may be performed prior to initiation of qualification testing on approval of the qualifying activity. If applicable, interchangeability of lighting colors shall also be demonstrated.

4.6.31 Reliability assurance tests. Reliability assurance tests shall be conducted using MIL-STD-781 for the qualification Phase and Sampling Phase tests. Altimeters selected for reliability assurance tests shall first be subjected to the group A, subgroup I inspection tests of table IX.

4.6.31.1 Reliability qualification phase. Prior to the acceptance of the altimeters under contract or order, three (3) to six (6) altimeters shall be tested as outlined in MIL-STD-781 under the section entitled "Qualification Phase of Production Reliability Tests" using test level E, except as specified herein. Criteria for test plan II shall be used. The altimeter shall demonstrate a minimum acceptable MTBF θ_1 of 1,000 hours as defined in MIL-STD-781.

4.6.31.1.1 Duty cycle. The duty cycle of test level E shall be modified as follows:

a. The altimeter shall be pneumatically cycled from sea level to approximately 45,000 feet altitude and return to sea level within approximately 15 minutes for at least 4 cycles per hour.

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During the external vibration period (10 minutes of each hour), the cycling rate shall be increased to 20,000 fpm. Period B of MIL-STD-781, figure 1 shall be approximately 4 hours. The barometric setting knob shall be rotated at least once throughout the complete range each 24 hour period.

b. During each hour the electric power to the integral vibrator shall be on for 57 minutes and off for 3 minutes. A two minute warmup period is permitted prior to initiating of cycling.

c. Indicator lighting shall be on 57 minutes and off 3 minutes each hour.

d. The barometric setting output potentiometer shall be electrically energized and loaded during this test (57 minutes on, 3 minutes off each hour).

e. During each hour, 10 minutes external vibration of ±0.5 g at any frequency between 20 and 60 Hz.

4.6.31.2 Reliability sampling phase tests. Samples of the altimeters shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Sampling Phase of Production Reliability Tests". For the sampling phase, test level E (as modified herein) shall be used. An alternative Production Reliability test may be specified by the procuring activity to be performed for a fixed period of 250 hours. When the alternative Production Reliability Test Plan is specified in the contract or purchase order, the contractor shall prepare such a plan for the procuring activity approval (see 6.2.2). The plan shall consider the specific altimeter design and be based upon the reliability test plans of more recent production contracts. Guidelines will be specified by the procuring activity. Such a plan must provide for the hold of a production lot until the test is complete.

4.6.31.3 Longevity tests. Unless otherwise specified by the contract or purchase order, a longevity test shall be performed in accordance with MIL-STD-781, test plan XXVIII and the duty cycle herein, except that 3,000 hours operating time shall be accumulated on each of two units. Two of the altimeters used during the reliability assurance tests shall be used for this test and all operating time accrued during reliability testing may be counted toward the longevity time. The altimeter performance checks may be performed at 500 hour intervals in lieu of 250 hour intervals after completion of 1000 hours of testing for each indicator. Data shall be retained for procuring activity approval (see 6.2.2).

4.6.31.4 Performance characteristics to be measured during longevity and reliability assurance tests. The altimeters shall be observed each working day for any obvious failures such as rough pointer movement, barometric potentiometer output, excessive scale errors, vibrator operation, or failure of the integral lighting. At the end of each 250 operating hours, the altimeter shall be subjected to the following tests:

a. Barometric setting scale, 4.6.3

b. Case leakage, 4.6.5

c. Scale error at room temperature, 4.6.9

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d. Friction, 4.6.7

e. Integral lighting 4.6.12

The integral vibrator shall be checked to ascertain proper starting characteristics during the static friction test of 4.6.7. At the completion of the reliability and longevity tests, the altimeters shall be subjected to the following test:

a. Examination of product, 4.6.1

b. Barometric setting scale, 4.6.3

c. Barometric setting knob functional test, 4.6.4

d. Case leakage, 4.6.5

e. Friction stop and jump, 4.6.6

f. Static friction, 4.6.7

g. Position error, 4.6.8

h. Altitude display scale error, 4.6.9

i. Hysteresis and after effect, 4.6.9.1

j. Lighting, 4.6.12

5. PACKAGING.

5.1 Packaging and preservation. The altimeter shall be packaged in accordance with MIL-STD-794, level A or C, as specified in contract or order (see 6.2.1). For level A packaging, the method of preservation shall be in accordance with MIL-P-116, method 1A-5 without preservation compound, using reusable metal containers.

5.2 Packing. The altimeter shall be packed in accordance with MIL-STD-794, level A, B or C as specified in contract or order (see 6.2.1).

5.2.1 Level A. The packaged altimeter shall be packed in a protective shipping carton conforming to PPP-B-636, class weather resistant, style optional, special requirements. Protective packing material such as polystyrene or other suitable material shall surround the altimeter to protect the entire unit including dial glass, bezel, and a protruding barometric setting knob from damage.

5.2.2 Level B. The packaged altimeter shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. The level C packing for altimeters shall conform to MIL-STD-794 requirements for this level.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used, whenever total quantities for

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shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Altimeters, packed as specified in 5.2.1, shall be unitized on pallets in comformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.2.4.2 Level B. Altimeters, packed as specified in 5.2.2, shall be unitized as specified in 5.2.4.1, except that the fiberboard caps shall be class domestic.

5.2.4.3 Level C. Altimeters, packed as specified in 5.2.3, shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B, and C. In addition to any special or other identification marking specified in the contract (see 6.2), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.3.

6. NOTES.

6.1 Intended use. The altimeters covered by this specification are intended for standby or emergency use in aircraft to indicate the pressure altitude of the aircraft above some reference point, assuming standard conditions of temperature and pressure. The altimeter also provides an electrical output as a function of the barometric pressure setting for use in correction of the primary altitude displayed by cathode ray tubes or head up displays.

6.2 Ordering data.

6.2.1 Acquisition requirements. Purchasers should exercise any desired options offered herein; and acquisition documents should specify the following:

a. Title, number and date of this specification

b. Altimeter type (see 1.2)

c. If other than standard parts are to be used (see 3.2.1.1)

d. Whether Reliability Qualification Phase tests are required (see 4.6.31.1)

e. Reliability Sampling Phase tests (see 4.6.31.2) (specify MIL-STD-781 or alternative test plan)

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f. Longevity Tests (see 4.6.31.3)

g. Maintainability demonstration (see 3.3.6 and 4.6.30)

h. Levels of preservation - packaging and packing required (see 5.1 and 5.2)

i. Warranty decal information, if applicable (see 3.11)

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are invoked, and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.2.2.3, 3.2.2.6	Deviations, Request for	UDI-E-23102A
4.5.1.2.1.1e, 4.6	Procedure, Manufacture Production Test	DI-T-4903
4.5.1.2.1.1e	Report, Manufacture Production Test	DI-T-4903
4.5.2.1.1	Report, Reliability (monthly)	DI-R-7035
4.6.31.2, 4.6	Procedure, (Alternate) Reliability Assurance Test	DI-R-7035
4.6.31.3	Report, Longevity Test	DI-T-4904
4.5.2.1.1, 4.5.2.1.5	Report, Failure	UDI-R-21141A
4.6.30, 4.6	Procedure, Maintainability Test	DI-T-4903

(Data item description related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Good as new. "Good as new" shall mean the altimeter has been operated less than 10 percent of specified MTBF operation. Prior to delivery, the altimeter shall be subjected to and meet all the requirement of a new altimeter, except the run-in test need not be performed.

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6.3.2 Refurbished. "Refurbished" shall mean the altimeter has been completely overhauled with all component parts meeting current parts standards; and prior to delivery the altimeter shall have been subjected to and met all requirements of a new altimeter.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Product List whether or not such products have actually been so listed by that date. Suppliers are directed to Provisions Governing Qualification SD-6 and are urged to arrange to have the products tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Department of the Navy, Washington, D.C., 20360, and information pertaining to qualification of products may be obtained from that activity.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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